

AMANUEL AYALEW

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SUMMARY

Software Engineer with experience in backend systems and growing expertise in Machine Learning, with a focus on research-driven development. Experience in building and improving scalable applications, Skilled in experimentation, performance analysis, and exploring new approaches to model design. Passionate about applying machine learning systems to solve real-world, data-driven problems.

EDUCATION

Bachelor of Science in Software Engineering
Addis Ababa University (AAiT)

2021 – 2026

PROFESSIONAL EXPERIENCE

Icog Labs

July 2025 – May 2026

ML Engineer Intern

- Conducted research on predictive coding-based transformer architectures as alternatives to backpropagation
- Worked on a tag recommendation model for the Mindplex platform, improving recommendation accuracy by 30% through evaluation of learning efficiency and model behavior
- Experimented with Quantum Variational Circuits (QVC), reducing model parameter count by 80% while maintaining performance

Alet Systems

April 2024 – December 2024

Software Engineer

- Tested and validated backend APIs to ensure system reliability and performance
- Debugged and resolved production level issues in backend systems, cutting response/resolution time by 40%
- Supported full-stack and mobile application maintenance

PROJECTS

Predictive Coding Transformer (Research Project)

- Explored predictive coding-based transformer architectures as an alternative learning paradigm to backpropagation
- Conducted experiments to evaluate convergence behavior, efficiency, and performance

Awaqi – Bilingual AI Tax Assistant

- Built a bilingual (Amharic/English) RAG assistant delivering cited, grounded answers to taxpayer questions from official tax proclamations Designed a hybrid retrieval pipeline (pgvector + keyword search, fused via RRF) with citations and confidence scoring to prevent hallucination
- Built "Awaqi Max," a ReAct agent (Gemini function calling) reasoning across the knowledge base and live web search for complex queries Automated document ingestion (PDF/HTML, OCR, chunking, embedding) and delivered the assistant via web chat, Telegram, and an admin dashboard

SKILLS

- Programming:** Python, TypeScript, JavaScript, Go
- Machine Learning & Data:** PyTorch, scikit-learn, pandas, NumPy
- Backend & Data Engineering:** REST APIs, FastAPI, Node.js, Express.js, PostgreSQL, SQL, Airflow, GraphQL
- Frontend:** Next.js
- Tools & Technologies:** Git, Linux

CERTIFICATIONS & ACHIEVEMENTS

- A2SV Generative AI for Africa Hackathon Finalist (2023)
- Machine Learning Certification – AAiT (2023)
- Finalist – Reboot the Earth Hackathon (2024)
- Machine Learning Specialization (DeepLearning AI)